



Amphiphilic polymers based on higher alkylene oxides Synthesis and characterization by different chromatographic techniques

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ABSTRACT

Homopolymers and block copolymers of higher epoxides (butene oxide and hexene oxide) are synthesized using 1-alkanols and polyethylene glycol monomethyl ether (PEG-MME) 1100 as initiators by anionic ring opening polymerization in bulk. Most of the samples were synthesized with controlled microwave heating in sealed vessels. Tri- and tetrablock copolymers with different repeat units in the individual blocks are synthesized by living polymerization with addition of the next monomer after complete consumption of the previous one. The products thus obtained are characterized using size exclusion chromatography (SEC), liquid chromatography under critical conditions (LCCC) and liquid adsorption chromatography (LAC).

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1. Introduction

Poly(alkylene oxide)s are in widespread use for various applications [1]. Their properties depend on the substituents of the oxirane ring: while poly(ethylene oxide)s (PEO) are soluble in water, only the lower poly(propylene oxide)s are water-soluble, and the polymers of higher epoxides – such as butene oxide (BO), hexene oxide (HO), etc. – are soluble only in non-polar organic solvents. Consequently, block copolymers containing ethylene oxide (EO) in the hydrophilic block and a higher epoxide in the hydrophobic block are amphiphilic materials, which can be used as surfactant or emulsifier. Up to now, only block copolymers of EO and propylene oxide (PO) are materials of commercial interest, which are produced in large amounts [2–4], while block copolymers containing higher alkylene oxides are still not in widespread use [5,6]. They may, however, have very special physical properties, as a block containing longer alkyl side groups is much more hydrophobic than a PO block. One can imagine, that even more interesting materials may be obtained by adding a third block with a different side chain: such A–B–C triblocks and A–B–C–D tetrablocks (where A is EO, B is PO,

C is BO, and D is HO) could be tailor-made surfactants for special applications.

In principle, one may synthesize tri- and tetrablocks by living anionic polymerization starting with the hydrophilic block (such as a PEG-monomethyl ether) or the hydrophobic block (such as a fatty alcohol) and add the next monomers after consumption of the previous one.

The problem in the evaluation of such materials is their characterization: as traces of the homopolymers (A, B, or C) and the diblock copolymers (A–B, B–C) may influence the physical properties very strongly, one must be sure to keep the concentration of these impurities as low as possible. The situation is even more complex with A–B–C–D tetrablocks.

In this study, analytical techniques have been developed, which allow the identification and determination of side products in the synthesis of A–B–C and A–B–C–D block copolymers.

2. Characterization of block copolymers

A–B diblock copolymers and triblocks containing just two different repeat units (A–B–A and B–A–B) can be characterized using different chromatographic techniques, as has already been shown in several publications [2,7–12,13].

The overall molar mass can be obtained by size exclusion chromatography (SEC), which separates according to the hydrodynamic volume. SEC can be performed with two different concentration

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Table 1
Possible situations in LCCC of amphiphilic block copolymers.

Column	Mobile phase polarity	Hydrophilic (EO)	Hydrophobic (PO, BO, HO)
RP	High	LCCC	LAC
RP	Low	SEC	LCCC
NP	High	LCCC	SEC
NP	Low	LAC	LCCC

detectors (such as refractive index and density [14–16]), the sensitivity of which for the repeat units is sufficiently different. Using just one concentration detector may result in serious errors, if the composition of the copolymer changes along the molar mass distribution (MMD).

SEC with dual detection yields information on the (average) chemical composition along the MMD [16–20], it is, however, not capable of discriminating block copolymers and a mixture of homopolymers.

A separation according to functionality or to the length of one block can be achieved by liquid chromatography under critical conditions (LCCC), which is performed in mixed mobile phases with a special composition and at a special temperature. At the so-called critical adsorption point (CAP) entropic and enthalpic effects compensate each other, and retention becomes independent on molar mass, which means, that the corresponding block becomes “chromatographically invisible”, and a separation according to other blocks or the end groups becomes possible. Depending on the chromatographic system, there are different options, as has been discussed previously [2,21] (See Table 1).

On reversed phase columns, there may be critical conditions for the hydrophilic block in a mobile phase with rather high polarity. In such a mobile phase the hydrophobic block will elute in LAC mode. On the other hand, critical conditions may also be found for the hydrophobic block in a rather non-polar mobile phase. In this case, the hydrophilic block will elute in SEC mode.

On normal phase columns, a CAP for the hydrophilic block may be found in a polar mobile phase, in which the hydrophobic block will elute in SEC mode. In a mobile phase with low polarity a CAP may exist for the hydrophobic block: in this case, the hydrophilic block will elute in LAC mode.

For the purpose of this study, the situation with an excluded second block (at the CAP for the first one) is more favorable, as it works almost for any length of the non-critical block.

Such conditions have already been realized previously for EO–PO block copolymers [22,23]. Obviously, at the CAP for PEO on a normal phase column higher polyalkylene oxides will also elute in SEC mode (as PPO). On the other hand, the more polar block (PEO or PPO) will elute in SEC mode on a reversed phase column at the CAP for a higher polyalkylene oxide. Hence the next step was to find critical conditions for PBO and PHO on a reversed phase column.

3. Experimental

3.1. Polymerizations

PO, BO, HO and *n*-alcohols (butanol, hexanol, dodecanol, octadecanol) were purchased from Fluka (Buchs, Switzerland).

The lower alcohols (butanol and hexanol) were dried over anhydrous sodium sulfate, while epoxides were refluxed over calcium hydride prior to use.

Dodecanol and octadecanol were used as received. Polyethylene glycol monomethyl ether (PEG-MME) with an average molar mass of 350 and 1100 (specification by the producer) was purchased from Fluka (Buchs, Switzerland). PEG-MME was dried by azeotropic distillation with toluene. Sodium hydride (60% dispersion in mineral oil) from Aldrich was used as received.

The polymerization of PO, BO and HO was performed by anionic ring opening using the different initiators (different alcohols and PEG-MME 1100) and sodium hydride as co-initiator. The details of the synthesis will be discussed in another publication [24].

All polymerizations were performed in 2–5 ml (filling volume) microwave process vials (Biotage AB), which were oven dried under vacuum prior to use. The initiator (alcohol or PEG-MME 1100) and sodium hydride were added to the vials which were subsequently sealed with an aluminum crimp top and Teflon septum. The vials were then evacuated and purged with argon (three cycles). The calculated volume of epoxide was added with a syringe through the septum. The quantity of sodium hydride was 0.1% of total product and quantities of initiator (alcohol or PEG-MME) and epoxide in different experiments depend upon the intended average number of monomer units in the chain (calculated by molar ratios).

The sealed microwave process vials were subsequently introduced into the cavity of the single-mode microwave reactor (Biotage Initiator Eight EXP 2.0, absorbance level: high). The reaction temperatures ranged from 160 to 200 °C. The total reaction time depended on the monomer and the reaction temperature. After the polymerization the active end was neutralized by adding an equimolar quantity of acetic acid. The time required for complete conversion was determined by measuring the monomer content, as will be described later on.

The same procedure was followed for the synthesis of amphiphilic multiblocks. After complete conversion of the first monomer the reaction vial was cooled down to approximately 100 °C and the next monomer was added by a syringe through the Teflon septum without unsealing the vial, which was again introduced into the cavity of the microwave reactor.

After polymerization of the last monomer the active end was neutralized by adding an equimolar quantity of acetic acid. Details of the procedure will be given elsewhere [24].

Tetrablock copolymer with PEG-MME 350 as hydrophilic block and PPO, PBO and PHO as second, third and fourth blocks was prepared by slow addition of monomer with conventional heating. 8.0 g of PEG-MME 350 and 0.08 g of Na metal was added in a reaction flask equipped with a reflux condenser and a dropping funnel. After azeotropic distillation with toluene, slow addition of PO (8.0 g) was started under reflux at 160 °C.

After complete conversion of PO (which took about 5 h), slow addition of BO (8.0 g) (under reflux) took about 6 h at 160 °C. After complete conversion of BO, 8.0 g of HO was added and heated overnight (12 h) at 160 °C. Finally, the active chain end was neutralized by an equimolar quantity of acetic acid.

3.2. Chromatography

These investigations were performed using a density detector (according to the mechanical oscillator principle) DDS70 (Chromtech, Graz, Austria) in all chromatographic systems. Data acquisition and processing was performed using the software CHROMA, which has been developed for the DDS70.

In SEC, we used a modular system consisting of a Gynkotec 300C pump, a VICI injector equipped with a 100 µl sample loop, two column selection valves Rheodyne 7060 (Rheodyne, Cotati, CA, USA), a density detection system DDS 70 (Chromtech) coupled with a Bischoff 8110 refractive index detector (Bischoff, Leonberg, Germany). SEC measurements were performed on a 600 mm × 7.8 mm Phenogel 100 Å column (Phenomenex, Torrance, CA, USA) or a 300 mm × 7.8 mm PLgel MIXED E column (Polymer Laboratories, Church Stretton, UK) in chloroform at a flow rate of 1.00 ml/min. Sample concentrations were 3.0–10.0 g/l.

The columns were calibrated with PPGs from Sigma–Aldrich and Fluka (Buchs, Switzerland). Molar mass distributions were calculated using CHROMA.

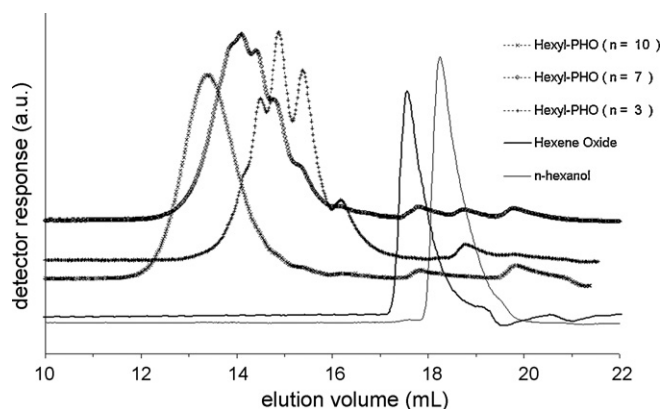


Fig. 1. SEC of poly(hexene oxides) initiated with n-hexanol (conventional heating), the monomer and the initiator. Detection: RI.

In LCCC, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA) at a flow rate of 0.5 ml/min. Samples were injected using an autosampler Spark SPH 125 Fix (Spark Holland, Emmen, The Netherlands) equipped with a 20 μ l loop. A refractive index detector RI 2414 (Waters) or a SEDEX 45 evaporative light scattering detector (ELSD) system (Sedere, France) was connected to the DDS 70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. Evaporator temperature: 30 °C.

LCCC was performed on the following columns:

Reversed phase: Jordi Gel DVB 500 RP (100% poly-divinylbenzene; 250 mm $\times$ 4.6 mm; particle diameter: 5 μ m; pore diameter: 500 Å, from Jordi, Bellingham, MA, USA).

Normal phase: Nucleosil 100-5 OH 5 μ m, silica-based diol phase; 250 mm $\times$ 4.6 mm; particle diameter: 5 μ m; pore diameter: 100 Å (Macherey-Nagel, Dueren, Germany).

In all systems, the columns and density cells were placed in a thermostatted box, in which a temperature of 25.0 °C was maintained for all measurements in LCCC and 30.0 °C in SEC. Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria).

The solvents – chloroform, tetrahydrofuran (THF), acetone and water, all HPLC grade – were purchased from Roth (Karlsruhe, Germany) and Merck (Darmstadt, Germany). PEG and PPG standards were purchased from Sigma–Aldrich and Fluka.

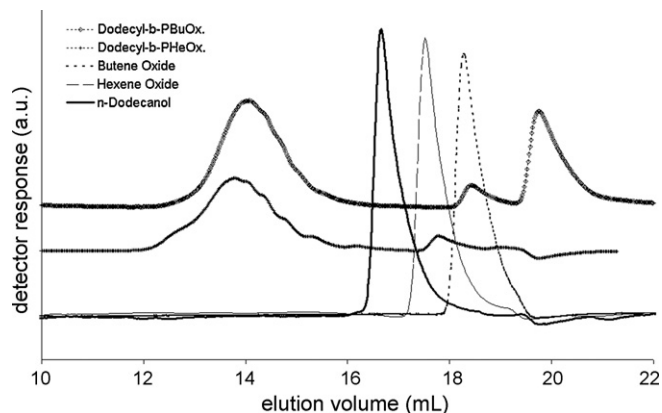


Fig. 2. SEC of poly(butene oxide) and poly(hexene oxide) initiated with n-dodecanol, the monomer and the initiator. Detection: RI.

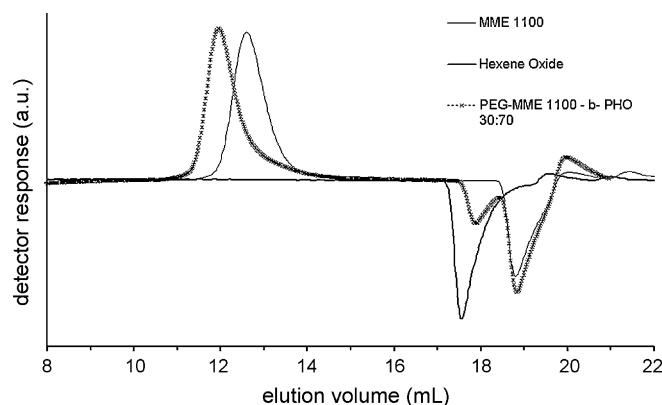


Fig. 3. SEC of a diblock copolymer of hexene oxide and PEG-MME 1100 (monomer-to-initiator ratio: 7:3), the monomer and the initiator. Detection: RI.

4. Results and discussion

Block copolymers can be synthesized by living anionic polymerization starting with functional chains as initiator, the end groups of which are activated by a catalyst. In this study, amphiphilic diblock copolymers were obtained by polymerization of a higher epoxide with PEG-MME as initiator.

For comparison, homopolymers of BO and HO were synthesized using the corresponding alcohol (n-butanol or n-hexanol) and more hydrophobic alcohols (dodecanol and octadecanol). These can also be considered as AB diblocks, where the block A (the alkyl chain) is monodisperse.

The first question concerns the conversion of the monomer and initiator, which is rather easy with PO: because of the low boiling point of PO, the internal pressure in the vial is an excellent measure for the presence of residual monomer [22,23].

As the vapor pressure of the higher alkylene oxide is considerably lower, a drop of the internal pressure to zero does not necessarily mean complete conversion.

In the case of BO and HO the residual monomer can be determined by SEC, as can be seen in the following figures.

Fig. 1 shows an overlay of chromatograms of three poly(hexene oxide) samples (synthesized with hexanol as initiator under conventional heating), the initiator and the monomer. Even when the corresponding alcohol (with the same number of carbon atoms as the monomer) is used as initiator, it can be sufficiently separated from the monomer. These chromatograms were obtained on the Phenogel 100 Å column, on which the alcohols and the monomer

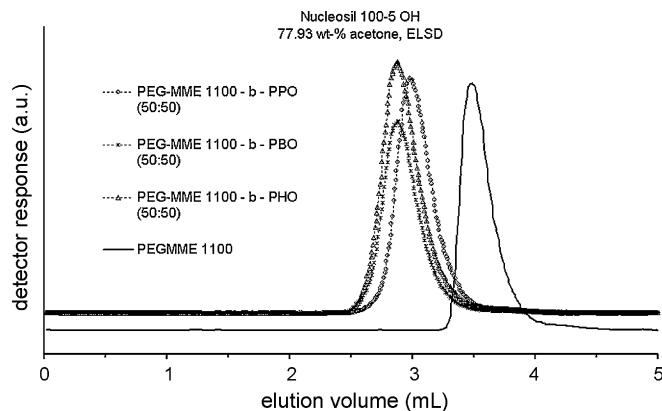


Fig. 4. LCCC of block copolymers of PEG-MME 1100 with PO, BO and HO, as obtained on Nucleosil 100-5 OH in 77.93% w/w acetone (CAP for EO). Detection: ELSD.

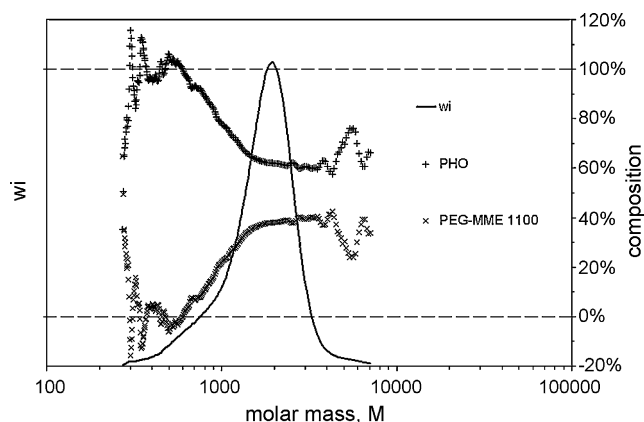


Fig. 5. MMD and chemical composition of the sample shown in Fig. 3, as obtained by SEC with coupled density and RI detection.

elute as quite unsymmetric peaks, which may be due to a mixed retention mode for these small molecules.

In one of the samples shown here the conversion of the monomer was almost complete, while the other one still contains some monomer and initiator. Both peaks can be reasonably integrated.

Fig. 2 shows chromatograms of a poly(butene oxide) (PBO) and a poly(hexene oxide) (PHO), the initiator (dodecanol), and the monomers. Dodecanol elutes clearly later than the lowest oligomers of BO and HO, but earlier than the monomers. Evidently, both polymers shown here still contain some residual monomer, but no initiator. The peak eluting at 20 ml is a solvent peak resulting from preferential solvation (the chloroform used contained 1% ethanol as a stabilizer).

In the case of a polymeric initiator (PEG-MME) the situation is even more favorable for the determination of residual monomer.

Fig. 3 shows an overlay of the chromatograms obtained by SEC with a diblock copolymer of hexene oxide initiated with PEG-MME 1100 (monomer-to-initiator ratio: 7:3), the initiator and the monomer.

The monomer (hexene oxide) elutes before the solvent peak (in this chromatogram THF was used as internal standard for flow rate correction) and can thus be quantified very well. As can be seen, this sample still contains considerable amounts of the monomer. The peaks of the initiator (PEG-MME 1100) and the block copolymer, however, overlap in SEC.

The information in the presence of residual initiator can be provided by LCCC, as has been shown previously [2]: at the CAP for

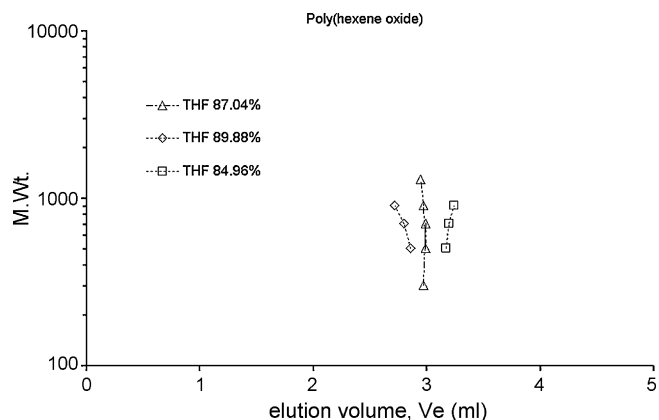


Fig. 7. LCCC: molar mass vs. V_e , as obtained for poly(hexene oxide) on the Jordi DVB 500A RP column in THF–water, flow rate 0.5 ml/min, detection: ELSD.

block A (in this case EO) on a normal phase column (NucleoSil 100-5 OH) in an acetone–water mobile phase (78%, w/w acetone) the initiator (PEG-MME) elutes close to the void volume, while the block copolymer as well as the homopolymer of the hydrophobic unit (PO) elute in SEC mode. The same system works also with the higher epoxides.

Under the same conditions, both poly(butene oxide) and poly(hexene oxide) elute in SEC mode and can thus be clearly separated from PEG (and PEG-MME). As can be seen in Fig. 4, all three block copolymers (containing different poly(alkylene oxides) as hydrophobic block) do not contain residual initiator.

The next question concerns the presence of the homopolymer of the second monomer (butene oxide or hexene oxide), which may be formed by a chain transfer reaction [1,6,22,25,26].

The diblock copolymer of hexene oxide with PEG-MME 1100 shown in Fig. 3 (monomer-to-initiator ratio: 7:3), contains not only considerable amounts of the monomer: the copolymer peak has a low molecular end, which may be due to residual initiator or the side product poly(hexene oxide).

As has already been discussed above, SEC with dual detection allows the determination of the chemical composition along the MMD [16,19]. The result is shown in Fig. 5: the low molecular fraction contains indeed approximately 100% poly(hexene oxide). While the overall composition of the sample agrees with the theoretical value (70%), the main fraction contains considerably less than the intended 70% hexene oxide, which is reasonable, as a considerable part of the monomer had not been incorporated in the copolymer.

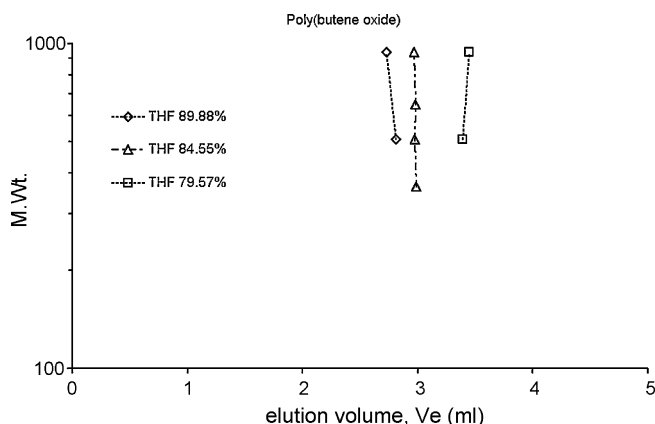


Fig. 6. LCCC: molar mass vs. V_e , as obtained for poly(butene oxide) on the Jordi DVB 500A RP column in THF–water, flow rate 0.5 ml/min, detection: ELSD.

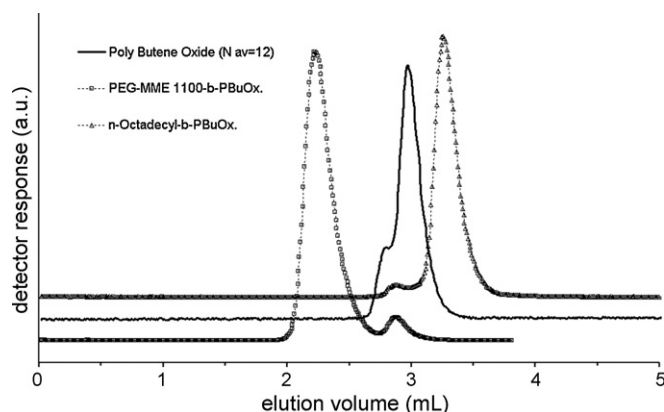


Fig. 8. LCCC of poly(butene oxide)s at the CAP for BO, as obtained on the Jordi column in 84.55%, w/w THF–water, flow rate 0.5 ml/min, detection: ELSD.

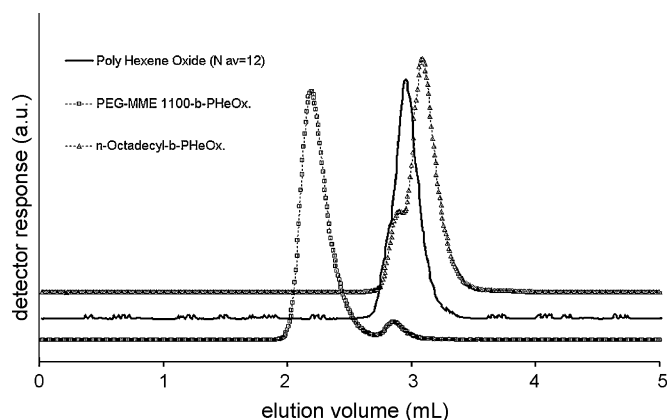


Fig. 9. LCCC of poly(hexene oxide)s at the CAP for HO, as obtained on the Jordi column in 87.04% (w/w) THF–water, flow rate 0.5 ml/min, detection: ELSD.

The average composition of the polymer peak means, however, that the fraction eluting in the corresponding interval contains a certain amount of the individual monomer units, but not, that it is really a copolymer. The result is only unambiguous, if a fraction contains close to 100% of a monomer unit.

The homopolymer of monomer B in an AB diblock copolymer can be determined at the CAP for B, under SEC conditions for A. If the block B is more hydrophobic than A, this may be the case on reversed phase columns.

As we have shown previously, critical conditions for PO could be realized on a Jordi Gel DVB 500 RP column in acetone–water (92.42%, w/w acetone) [2] or in THF–water (80.22%, w/w THF) [22,23].

Under these conditions, PEG elutes in SEC mode. Moreover, PPG diols (from chain transfer to water) and allyl-PPG (from chain transfer to the monomer) could not only be separated from the block copolymer, but also from each other [22,23]. We have also found critical conditions for poly(butene oxide) and poly(hexene oxide) on the same column in a comparatively more non-polar mobile phase composition. As can be seen, a CAP for BO is observed on the Jordi column in 84.55%, w/w THF (Fig. 6), for HO on the same column in 87.04%, w/w THF (Fig. 7).

Under these conditions, the block copolymer with PEG-MME can be separated from the homopolymers. In such mobile phases, the individual polymer homologous series of longer epoxides (diols and alkenyl-terminated chains) cannot be separated from each other, as these end groups are not sufficiently different.

In LCCC of a block copolymer there can be two possible situations: at the CAP for one block, the second block could be excluded

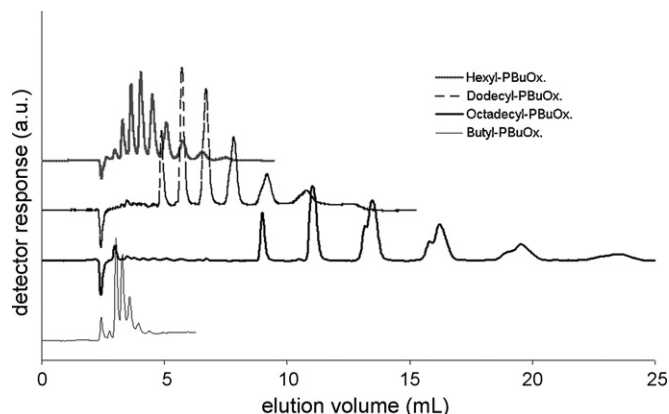


Fig. 10. Overlay of LAC chromatograms of poly(butene oxide)s with different end groups. Prodigy ODS3, 90.06%, w/w acetone–water, RI detection.

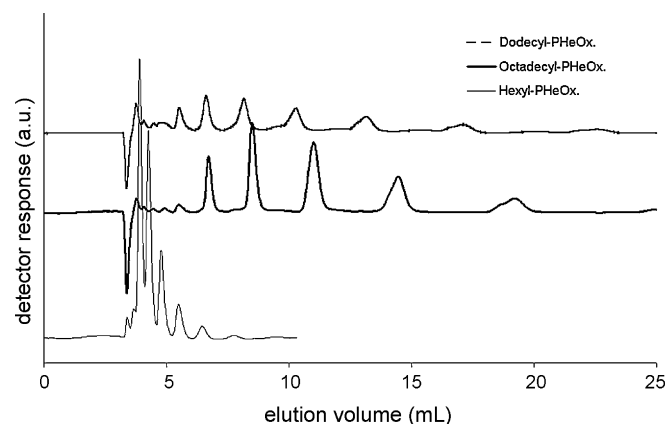


Fig. 11. Overlay of LAC chromatograms of poly(hexene oxide)s with different end groups. Spherisorb S5XC18, 90.06 wt% acetone–water, RI detection.

from the pores (PEG) or adsorbed more strongly on the surface of the stationary phase. If the second block elutes in the exclusion regime, the copolymer will elute earlier than the homopolymer, if it is adsorbed, it will elute later.

In the first case the second block must be large enough to be separated from the homopolymer. This is the case for polymeric initiators, such as PEG-MME 1100.

The second case is more favorable for functional chains containing a (monodisperse) adsorbing end group, the interaction of this group with the surface should be strong enough to separate the functional chains from the non-functional ones.

In this study we used both kinds of block copolymers: with a block in exclusion regime (PEG) and with a strongly adsorbing block (the dodecyl or octadecyl group).

In the case of poly(butene oxide), both situations are quite favorable and we were able to separate the block copolymers (whether with PEG-MME 1100 or octadecanol) from the homopolymer (which may be formed by chain transfer to monomer) (see Fig. 8).

In the case of poly(hexene oxide), the polymer itself is highly non-polar, hence a more adsorbing end group has not much an effect, but appears only as a shoulder, if the sample contains a considerable amount of homopolymer. Of course the situation with an excluded second block (PEG-1100 monomethyl ether) is more favorable in this case (Fig. 9).

There is, however, one more situation, which can be utilized in the characterization of AB diblocks, where the block A is monodisperse (such as an alkyl end group of a monofunctional chain): in the low molecular range, the individual oligomers may be completely

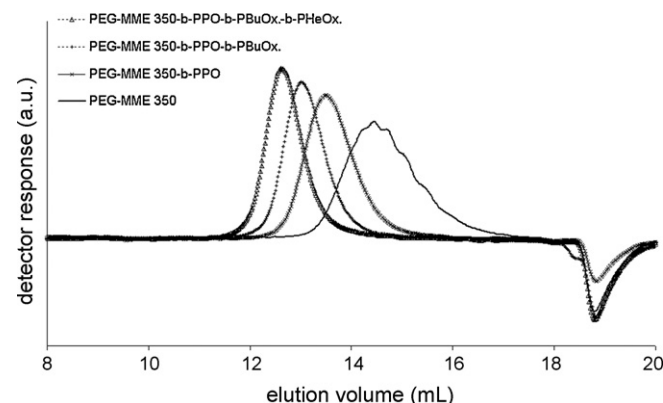


Fig. 12. SEC of block copolymers obtained by anionic polymerization of alkylene oxides with conventional heating.

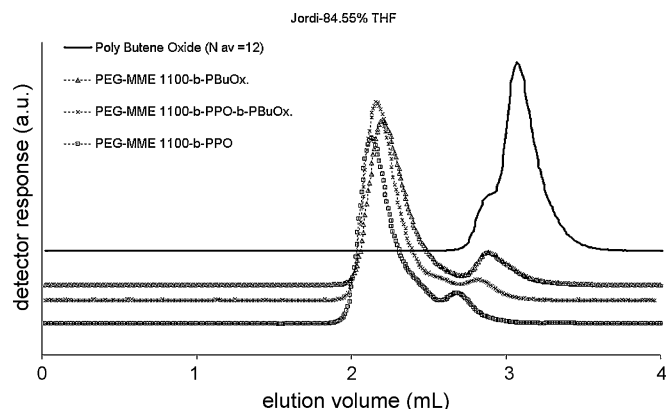


Fig. 13. LCCC at the CAP for BO, as obtained on the Jordi column in 84.55%, w/w THF–water, flow rate 0.5 ml/min, detection: ELSD.

resolved by LAC (at a positive interaction parameter c_B of the repeat unit). If the end group is adsorbed more strongly than the repeat unit (the interaction parameter of A is larger than that of B), a second series of peaks will appear in LAC.

This could be achieved on the Prodigy ODS3 and the Spherisorb S5XC18 column in acetone–water mobile phases with about 80–95%, w/w acetone.

Fig. 10 shows such a LAC separation of poly(butene oxide)s with the same average degree of polymerization ($n_{av} = 4$), but different end groups, which was obtained on the Prodigy ODS3 column in 90.04% acetone. With increasing hydrophobicity of the end group, retention (and selectivity) increases dramatically. The homopolymer (with butyl or butenyl end groups) elutes between 3 and 5 ml. In the polymers with a dodecyl or octadecyl end group it should be clearly visible. Evidently, the dodecyl PBO seem to contain traces of the homopolymer. The nature of the shoulder in the higher oligomers of the octadecyl PBO is not yet clear. One may speculate, that it could be due to different sequences (head to head or head to tail).

Fig. 11 shows a LAC separation of poly(hexene oxide)s with different alkyl end groups and approximately the same average degree of polymerization ($n_{av} = 4$), on the Spherisorb S5XC18 column in 90.04% acetone.

While the characterization of A–B diblocks appears to be solved by the techniques described above, the situation is much more difficult with tri- and tetrablocks (A–B–C and A–B–C–D).

Such polymers with PEG as hydrophilic block and hydrophobic blocks with increasing hydrophobicity (from PO to BO and HO) may have highly interesting physical properties. These polymers may

be synthesized by anionic living polymerization with subsequent addition of monomers. Obviously, at each step of the polymerization a high purity of the intermediate di- and triblock is inevitable.

Fig. 12 shows an overlay of size exclusion chromatograms obtained with PEG-MME 350, the corresponding diblock with PO, the triblock obtained therefrom by addition of BO, and the tetrablock with HO as the last block. These samples were obtained by subsequent ring opening polymerization of PO (with PEG-MME as initiator), BO and HO in the following amounts: PEG-MME 350: 8 g, PO: 8 g, BO: 8 g, HO: 8 g. As one may expect, the chain transfer reaction to the monomer should result in rather short chains of the homopolymers, they should appear as a low molecular shoulder (at higher elution volume), which is obviously not the case in these samples.

Apparently, dual detection cannot be applied in an A–B–C triblock, as the A–B diblock is polydisperse in both blocks (A and B), consequently, one could only determine an average response factor.

There is, however, a much better option: if the A–B diblock is sufficiently pure (which means, it does not contain residual initiator A and homopolymer B), the only important side product in the A–B–C triblock will be the homopolymer C. Under appropriate reaction conditions, complete conversion of the monomer and the initiator (A–B) should be achieved.

At the CAP for the block C, the homopolymers A and B as well as the diblock A–B should elute in SEC mode, which would allow to separate them from C.

This is indeed the case, as can be seen in Fig. 13: at the CAP for BO, the first block (PEG-MME) and possible diblocks (PEG-MME–PO and PEG-MME–BO) are well separated from the BO homopolymer. A shoulder in the chromatograms of the diblocks indicates the presence of some homopolymer.

The same approach can also be applied to the tetra-block, as can be seen in Fig. 14: at the CAP for HO, all other structures elute in SEC mode and are thus clearly separated from the HO homopolymer.

It must be mentioned, that the di- and triblocks shown here were not the precursors for the tetrablock. We have just selected bad samples, in which the homopolymer can be clearly seen. Under appropriate polymerization conditions, which are described in a different publication [24], the intermediates do not contain a measurable quantity of the homopolymers.

5. Conclusion

Size exclusion chromatography with density and RI detection along with LCCC is an excellent tool for characterization of epoxide-based block copolymers. In addition it provides information on the homopolymer content in multiblock copolymers of epoxides at each step of polymerization.

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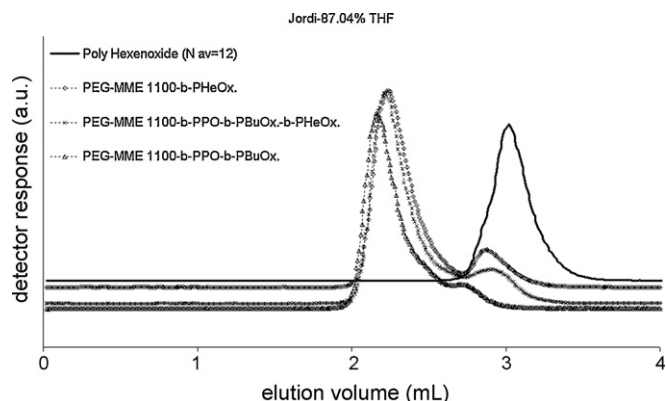


Fig. 14. LCCC at the CAP for HO, as obtained on the Jordi column in 87.04%, w/w THF–water, flow rate 0.5 ml/min, detection: ELSD.

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